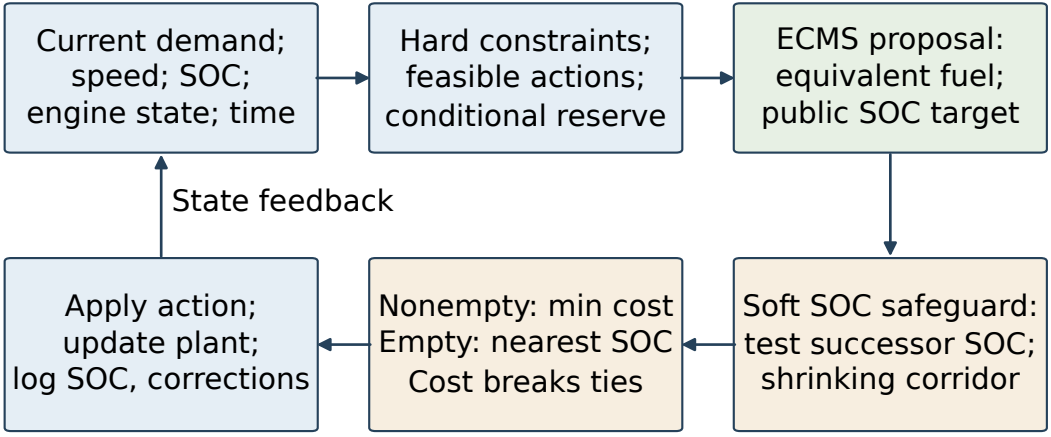
Current demand; speed; SOC; engine state; time
Hard constraints; feasible actions; conditional reserve
ECMS proposal: equivalent fuel; public SOC target
State feedback
Apply action; update plant; log SOC, corrections
Nonempty: min cost
Empty: nearest SOC
Cost breaks ties
Soft SOC safeguard: test successor SOC; shrinking corridor
Hard-feasible set empty: record failure and stop the virtual task